

# How Far Can Heuristic Learning Go? Characterizing Performance Ceilings, Complexity, and Sample Efficiency in Large-Scale Real-World Competitive Games

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This paper investigates the fundamental capabilities and limitations of heuristic learning (HL) in complex, large-scale real-world competitive games. We establish a comprehensive benchmark and characterize the performance ceilings, computational complexity, and sample efficiency of HL approaches across diverse game domains.



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